TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Charge Planning Tool (CPT) for Fleet-aware Smart Charging of Electric Vehicle Fleets under Grid Constraints			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-generation Energy (), Bio-health ( ), Aerospace ( ), Quantum ( ), Advanced Robotics & Manufacturing ( ), AI (), Cybersecurity (), Hydrogen ( ), Semiconductor materials ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communications		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Internet SW	Heavy Electrical Equipment	Power conversion equipment
Concept of Proposed Technology	**Overview** TNO proposes to deploy and further develop its Charge Planning Tool (CPT) to enable fleet-aware smart charging at depots and charging hubs where the number of vehicles exceeds the number of chargers and the grid connection is limited. The CPT is designed to integrate with existing Fleet Management Software (FMS), taking the logistics plan as input and producing an			

optimal charging plan that respects operational timing and energy requirements. [1]

Key technical capabilities (current CPT baseline)

Charging schedule optimisation for a non-homogeneous fleet with multiple charging opportunities, fewer chargers than vehicles, charger-vehicle compatibility constraints, and grid-related constraints. [1]

Cost-aware optimisation that can minimise charging-operation cost components (for example electricity price components and battery-related costs), while maintaining feasibility and robustness of the fleet logistics plan. [2]

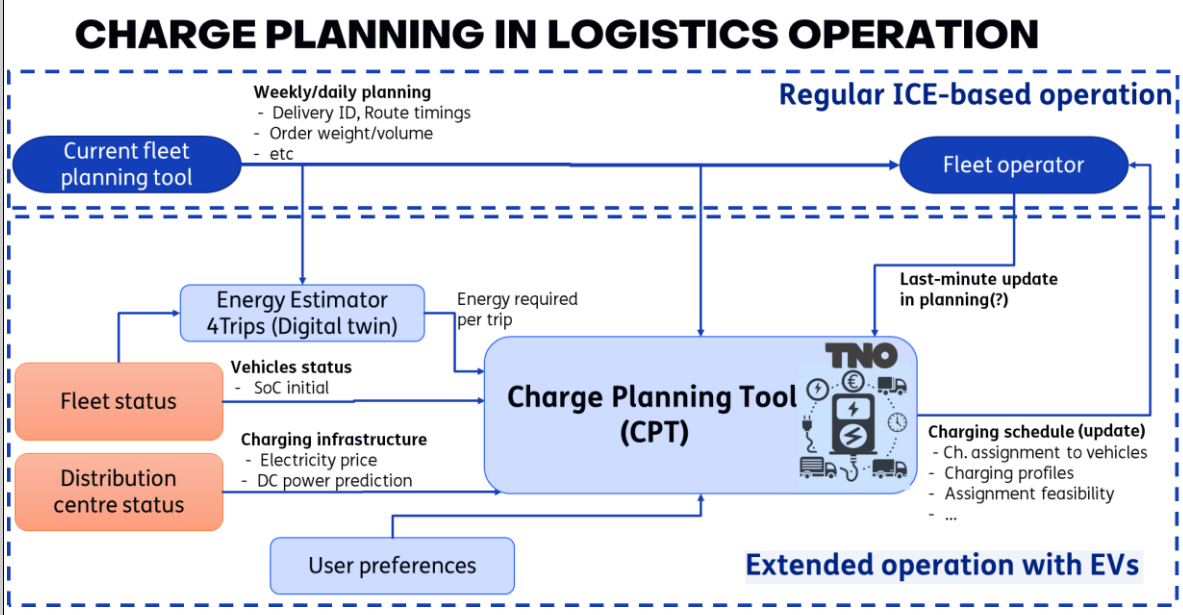
Smart charging at a single depot or similar facility: charger allocation to vehicles over time with a defined charging profile per session, based on vehicle arrival/departure times (ETA/ETD) and required trip energy. [4]

Two-stage optimisation architecture as demonstrated in the CPT literature: (i) a fast initialisation heuristic inspired by Multi-Processor Scheduling Problems (MSP) to rapidly construct feasible schedules and (ii) a Genetic Algorithm to improve schedules, with parallelised operations for computational efficiency. [4]

Battery-degradation-aware scheduling methods validated in TNO publications, including calendar and cyclic ageing considerations and grid limitations, with demonstrated real-time execution for large fleets in simulation. [3]

Optional extension in this joint R&D (subject to partner requirements)

Integration with battery digital twins to improve prediction of charging power profiles and battery ageing, supporting more robust schedules and better infrastructure planning. [4][5]



Rationale for Joint R&D Project	Logistics operators and charging infrastructure providers increasingly face limited or constrained grid connections, which prevents simultaneous charging and creates a planning bottleneck. TNO has highlighted that charging time must be integrated into logistics planning and that existing fleet planning tools for combustion vehicles do not provide a robust solution for this. [2] **This joint R&D project will combine:** TNO's CPT algorithms, charging schedule optimisation expertise, and battery modelling and digital twin methods as published in peer reviewed and conference literature. [1][3][4] The Korean SME partner's operational know-how in charging hardware and infrastructure, and access to real operational datasets and pilot sites for validation. The collaboration is needed to adapt and validate CPT in a Korean operational environment, including local fleet use cases, charging hub configurations, and business constraints, and to accelerate industrial deployment in Korea.
Key Words	1. Fleet smart charging 2. Charge planning 3. Grid constraint optimization 4. Battery degradation aware scheduling 5. Battery digital twin

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (5)	End of the Project	Level (7)
R&D Objectives	Integrate TNO Charge Planning Tool with the Korean SME partner's charging operations platform (or FMS interface), using an API-based data exchange of routes/trips, vehicle constraints, charger assets and grid limits. [1][2] Extend CPT optimisation to cover Korean pilot use cases (e.g., depot fleets and high power charging hubs) while maintaining real-time feasibility for day-to-day operations. [2][3] Incorporate battery-degradation-aware cost terms (calendar and cyclic ageing) and validate their impact on charging cost and operational robustness using fleet data and simulation. [3][4] Demonstrate (pilot) measurable improvements in charger utilisation and peak power management under a limited grid connection, and provide decision support for charger and grid upgrade investments. [2]			
R&D Target Specifications	**Functional targets** Support heterogeneous fleets and chargers (different max powers) and explicit charger-vehicle compatibility constraints. [1][4] Constraints: ETA/ETD time windows per vehicle, required trip energy, limited number of chargers, and contractual grid capacity. [1][3][4] **Performance targets (project targets, to be validated in pilots)** Schedule computation time: less than 180 s for a representative fleet size of up to 200 vehicles (baseline demonstrated less than 160 s for 200 vehicles in simulation). [3] Cost impact: demonstrate reduction in total charging operation cost relative to baseline scheduling; baseline literature reports 13% savings in a realistic simulation case study. [3] Robustness: quantify schedule feasibility under timing uncertainty (late arrival, higher energy consumption) and evaluate mitigation via safe margins and, if required, digital twin based prediction improvements. [2][4]			
R&D Plan	**Year 1: Requirements, data interfaces, and baseline CPT integration** Define pilot use cases (fleet types, depot/hub topology, charger types, grid limits) and data interfaces with partner systems. Implement secure data ingestion pipelines and baseline CPT schedule generation for the			

selected use cases. [1][4]

Year 2: Optimisation extensions and validation

Implement cost components relevant to the pilot (electricity price profiles, peak power penalties, battery degradation costs). [2][3]

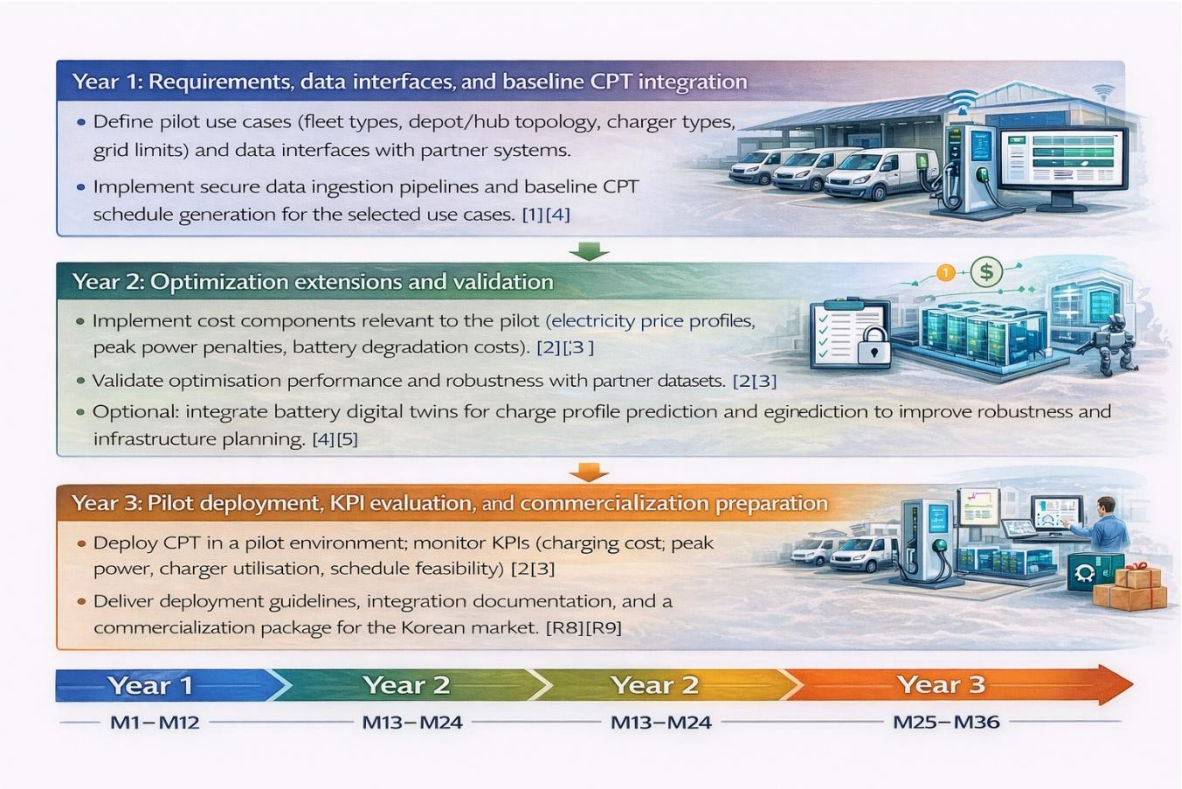
Validate optimisation performance and robustness with partner datasets; iterate based on operational feedback.

Optional: integrate battery digital twins for charge profile prediction and ageing prediction to improve robustness and infrastructure planning. [4][5]

Year 3: Pilot deployment, KPI evaluation, and commercialization preparation

Deploy CPT in a pilot environment, monitor KPIs (charging cost, peak power, charger utilisation, schedule feasibility).

Deliver deployment guidelines, integration documentation, and a commercialization package for the Korean market.



4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Provide CPT core algorithms and software modules; configure optimisation problem definition for the pilot scenarios. [1][4] Provide battery and charging modelling expertise, including battery-degradation-aware cost modelling and, if selected, battery digital twin integration. [3][4][5] Perform simulation-based verification and prepare validation protocols; support on-site pilot commissioning and performance assessment. Lead dissemination of scientific results (publications) and jointly define IP and exploitation plan with the partner.
Expected R&R of Korean SME Partner	Provide operational requirements and datasets: charging hub/depot layouts, charger specifications, grid connection limits, and (where applicable) fleet schedule inputs and measured charging sessions. Provide system integration support (interfaces to charger management, energy management and/or fleet operations platforms) and host pilot demonstrations in Korea. Support evaluation of KPIs and commercialization planning for the Korean market, including customer engagement and deployment scaling.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical outcomes** Integrated CPT prototype for the Korean pilot use case(s), producing feasible depot or hub charging schedules under grid and asset constraints. [1][2][4] Validated optimisation performance and battery-degradation-aware cost modelling, with documented KPI improvements versus baseline scheduling. [3] **Business and industrial impact** Higher utilisation of existing charging assets and reduced peak power draw, supporting operation under limited grid connections and potentially deferring costly grid upgrades. [2] A software differentiator for Korean charging infrastructure providers and fleet operators: charge planning as a service layer that can be integrated with existing FMS and charger management systems. [1][2]

6. References

[1] A. Beckers, R. Medina, S. Wilkins, "Charge Planning Tool for Heavy-Duty Electric Vehicle Fleets", conference paper, 2024, TNO Repository (UUID: 4043c8dc-e109-4441-8b23-0ea3a2fddf68). URL: https://resolver.tno.nl/uuid:4043c8dc-e109-4441-8b23-0ea3a2fddf68 (accessed 19 Dec 2025).

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[3] S. Rath, R. Medina, S. Wilkins, "Real-time Optimal Charging Strategy for a Fleet of Electric Vehicles Minimizing Battery Degradation", 2023 IEEE 3rd International Conference on Sustainable Energy and Future Electric Transportation (SEFET), DOI: 10.1109/SEFET57834.2023.10245996. PDF (TNO Publications): https://publications.tno.nl/publication/34641555/EemswK/rath-2023-real.pdf (accessed 19 Dec 2025).

[4] S. Rath, A. Beckers, P. Netto, R. Medina, S. Wilkins, "Charge Scheduling for a Fleet of Electric Vehicles using Battery Digital Twin", EVS38 Proceedings, June 2025. URL: https://evs38-program.org/images/Proceedings/D%20Charging%20Infrastructure%20and%20grid%20integration/197_Charge%20Scheduling%20for%20a%20Fleet%20of%20Electric%20Vehicles%20using%20Battery%20Digital%20Twin.pdf (accessed 19 Dec 2025).

[5] TNO, "Charging Energy Hubs (CEH): transition towards sustainable mobility", webpage. URL: https://www.tno.nl/en/sustainable/mobility-logistics/energy-management/charging-energy-hubs/ (accessed 19 Dec

2025).

[6] B. Kin et al., "Different Charging Strategies for Electric Vehicle Fleets in Urban Freight Transport", 2021. PDF (TNO Publications): https://publications.tno.nl/publication/34638914/JWAd0r/kin-2021-different.pdf (accessed 19 Dec 2025).

[7] Internal partner mapping: "KOREAN_BATTERY_SMEs.xlsx" (Korean charging infrastructure SME partner profile; charge planning tool and vehicle digital twins marked as potential), provided by project team, accessed 19 Dec 2025.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization